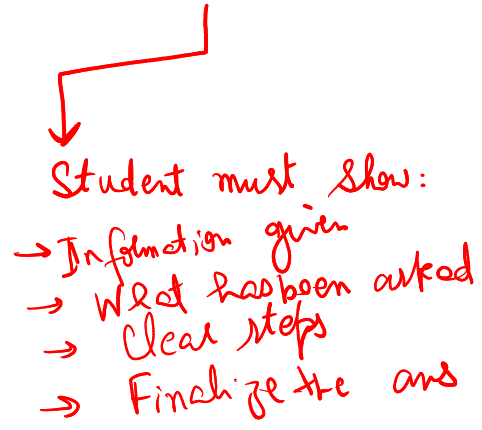


C SMC



Statement : *A statement is a sentence that is either true or false.*

$3 > 2$ True

$2 > 3$ False

The Truth of It All

Proof:

A proof is a convincing argument expressed in the language of mathematics that a statement is true.

Not guessing

All the primes are odd (False)
3, 5, 7, 11, 17, 19

[2]

Format of Proof

- Given:
- To prove / To find :
- Solution (steps)
- Therefore

Example: Prove that the sum of two even numbers is even.

Given: Let there two even numbers
 $2a, 2b$ $a, b \in \mathbb{Z}$

To Prove: Sum of $2a$ and $2b$ is even

Solution steps:
 $2a + 2b$
 $= 2(a + b)$

$a, b \in \mathbb{Z}$ so $a + b \in \mathbb{Z}$

$2(a + b)$ is divisible by 2 which makes
 $2(a + b)$ an even number

Therefore:

The sum of two even numbers is even

$\boxed{1}, 2, \boxed{3}, 4, \boxed{5}, 6, 7, \dots$

1. Prove that sum of two odd numbers is even.

Given: Let we have two odd numbers

$2a + 1$ and $2b + 1$

To Prove: Sum of $2a + 1$ and $2b + 1$ is even

Solution Steps:

$$2a + 1 + 2b + 1$$

$$= 2a + 2b + 2$$

$$= 2(a + b + 1)$$

(Distributive Property)

$a + b + 1 \in \mathbb{Z}$ and $2(a + b + 1)$ is divisible by 2

$2(a + b + 1)$ is even

Therefore

Remains

Sum of two odd numbers is even

Conditional Statement "If A, then B"

OR

A implies B

$$A \Rightarrow B$$

Hypothesis

Conclusion

Exp : If you study hard, then you will get a good grade.

Hypothesis

Conclusion

A	B	If A, then B
T	T	T
T	F	F
F	T	T
F	F	T

B	A	$B \Rightarrow A$
F	T	T

If $x > 2$ then $x^2 > 4$

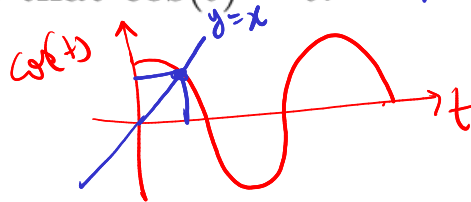
A	B	Exp	$A \Rightarrow B$
T	T	$x = 3$ Impossible	T Can not occur
T	F		
F	T	$x = -4$	T
F	F	$x = -1$	T

1. Two parallel lines in a plane have the same slope. T

2. $1 = 0$. F

3. The real number $x \not> 0$ (x is not greater than 0).

4. There is an angle t such that $\cos(t) = t$. T



^W1.1 Which of the following are mathematical statements?

✓ a. $ax^2 + bx + c = 0$.

✗ b. $(-b + \sqrt{b^2 - 4ac})/(2a)$.

✓ c. Triangle XYZ is similar to triangle RST .

✗ d. $3 + n + n^2$.

✓ e. For every angle t , $\sin^2(t) + \cos^2(t) = 1$.

1.2 Which of the following are mathematical statements?

✓ a. There is an even integer n that, when divided by 2, is odd.

✗ b. {integers n such that n is even}.

✓ c. If x is a positive real number, then $\log_{10}(x) > 0$.

✓ d. $\sin(\pi/2) < \sin(\pi/4)$.

1.4 For each of the following problems, identify the hypothesis (what you can assume is true) and the conclusion (what you are trying to show is true).

- a. r is irrational if r is a real number that satisfies $r^2 = 2$.
b. If p and q are positive real numbers with $\sqrt{pq} \neq (p+q)/2$, then $p \neq q$.
c. Let $f(x) = 2^{-x}$ for any real number x . Then $f(x) = x$ for some real number x with $0 \leq x \leq 1$.

Hyp: $f(x) = 2^{-x}$ for any real number x
Conclusion: There exist some real numbers x
such that $0 \leq x \leq 1$ and $f(x) = x$

^W1.9 Determine the conditions on the hypothesis A and conclusion B under which the following statements are true and false and give your reason.

a. If $2 > 7$, then $1 > 3$.

A B
F F T

b. If $x = 3$, then $1 < 2$.

T T True
F T True

^W1.13 Using Table 1.1, prepare a truth table for " A implies $(B$ implies $C)$."

A	B	C	$B \Rightarrow C$	$A \Rightarrow (B \Rightarrow C)$
F	F	F	T	T
F	F	T	T	T
F	T	F	F	T
F	T	T	T	T
T	F	F	T	T
T	F	T	T	T
T	T	F	F	F
T	T	T	T	T

^W**1.16** Suppose you want to show that $A \Rightarrow B$ is false. According to Table 1.1, how should you do this? What should you try to show about the truth of A and B ? (Doing this is referred to as a **counterexample** to $A \Rightarrow B$.)

^W**1.17** Apply your answer to Exercise 1.16 to show that each of the following statements is false by constructing a counterexample.

a. If $x > 0$, then $\log_{10}(x) > 0$. $0 < x \leq 1$

b. If n is a positive integer, then $n^3 \geq n!$ (where $n! = n(n-1) \cdots 1$).

n	n^3	$n!$	$n^3 \geq n!$
1	1	1	T
2	8	2	T
3	27	6	T
4	64	24	T
5	125	120	T
6	216	720	F
7	343	5040	F

$$n \geq 6$$

Home Work

- 1. Prove that sum of two odd numbers is even.*
- 2. Prove that 3999991 is not a prime number.*
(A prime number is a positive integer greater than 1 whose only positive divisors are 1 and itself.
For example, 7 is a prime number.) [CSMC 2021]
- 3. Exercise Questions from Chapter 1 (Solow)*

$$16 = 2 \times 8 \\ \quad \quad \quad \sim 4 \times 4$$

$$9 : 1, 3, 9 \\ 7 : 1, 7$$

$$7 = 1 \times 7 \\ 9 = 3 \times 3 \checkmark \\ \quad \quad \quad 1 \times 9$$

2. Prove that 3999991 is not a prime number.

(A prime number is a positive integer greater than 1 whose only positive divisors are 1 and itself.

For example, 7 is a prime number.) [CSMC 2021]

↓

3

Mon, Wed, Fri

✓

7 AM to 8 AM
(PST)

MONTH 1

Proof Writing + Algebra

- Solow: How to Read & Do Proofs — Ch 1–6 (all)
- AoPS Intermediate Algebra — Ch 4–6
(lines, parabolas, quadratic systems)
- Daily: write 1 proof by hand, no notes

★ Goal: solve B Q1 cleanly · target +3 pts on B Q1